Coefficient Stability and Interpretation in Credit Default Models with Correlated Predictors

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1 Introduction

Credit default modeling typically relies on a combination of financial and behavioral covariates that proxy for borrower risk. Financial variables such as credit limits, balances, and payment amounts capture exposure and utilization, while behavioral variables such as repayment history provide a direct measure of delinquency. Although these variables are often included jointly in credit risk models, they are not independent sources of information, but instead reflect overlapping aspects of borrower quality.

This dependence complicates statistical interpretation. Consider the linear projection

$$Y_i = X_i^\top \beta + \varepsilon_i, \tag{1}$$

where Y_i denotes the default indicator and X_i includes financial and behavioral covariates. The coefficient β_j represents the contribution of X_{ij} to the best linear predictor of Y_i conditional on the remaining covariates. When predictors are highly correlated, this contribution is identified only from variation orthogonal to the other covariates, making it sensitive to model specification.

This project studies how model interpretation changes in the presence of correlated predictors in a credit risk setting. The analysis focuses on whether repayment behavior explains most of the conditional variation in default, whether financial variables add independent explanatory power, and whether detailed repayment histories can be replaced with lower-dimensional summaries.

The emphasis is therefore on comparing information sets rather than validating a specific parametric model, which informs both the modeling approach and the interpretation of results.

2 Data

The analysis uses the UCI Default of Credit Card Clients dataset, which contains 30,000 observations. The response variable is a binary indicator of credit default in the following month.

Predictors are grouped into financial variables, behavioral variables, and demographic variables. The repayment variables are temporally ordered, with PAY_0 corresponding to the most recent repayment status.

A key modeling decision concerns how to represent repayment behavior. The raw repayment variables provide a high-dimensional and highly correlated description of delinquency across time. Including all such variables directly introduces redundancy and complicates interpretation.

Instead, we construct summary behavioral variables:

$$\text{ANY_DELAY}_i = \mathbf{1}\left(\max_t \text{PAY}_{it} > 0\right), \tag{2}$$

$$\text{MAX_DELAY}_i = \max_t \text{PAY}_{it}, \tag{3}$$

$$\text{NUM_DELAY_MONTHS}_i = \sum_t \mathbf{1}\left(\text{PAY}_{it} > 0\right). \tag{4}$$

These features capture key aspects of repayment behavior while reducing dimensionality. This choice is motivated by both statistical considerations, as highly correlated predictors lead to unstable coefficient estimates, and practical considerations, as credit risk models often rely on aggregated delinquency measures.

3 Exploratory Data Analysis

Our exploratory data analysis is conducted as an iterative process in which initial findings guide subsequent investigation. Rather than following a fixed sequence of predefined diagnostics, we begin with simple summaries and progressively refine the analysis based on observed structure in the data.

We first examine the relationship between repayment behavior and default outcomes. Figure 1 shows default rates as a function of the most recent repayment status (PAY_0). The pattern is strongly monotonic, with default rates increasing sharply as repayment status worsens, indicating that delinquency captures a large portion of variation in default outcomes.

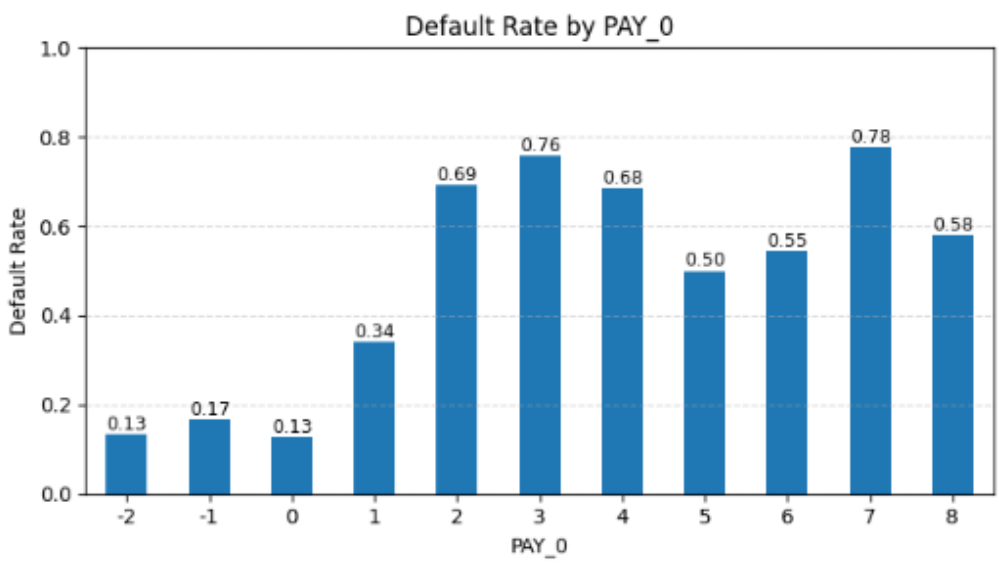

Figure 1: Default Rate per Number of Missed Payments in the last Period

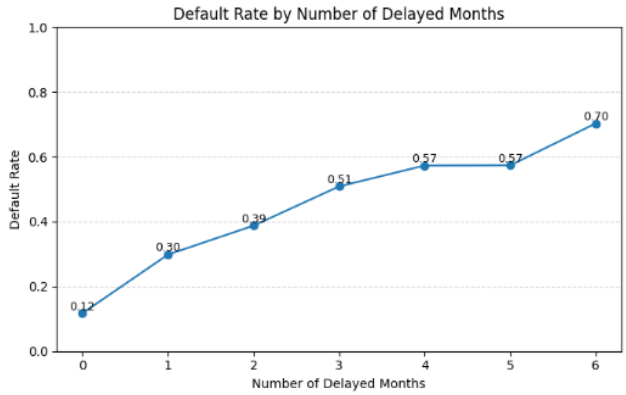

Figure 2: Default Rate by Number of Delayed Months

This finding motivates further examination of repayment behavior across time. The repayment variables exhibit similar patterns but are also highly correlated, suggesting redundancy. To assess whether simpler representations capture the same information, we construct summary measures of delinquency. Figure 2 shows default rates as a function of the number of delayed months, which exhibits a clear increasing relationship.

We next examine financial variables. Figure 3 shows default rates across credit limit deciles. While higher credit limits are associated with lower default rates, the relationship is more gradual compared to repayment behavior, indicating that financial variables provide a broader signal of borrower condition.

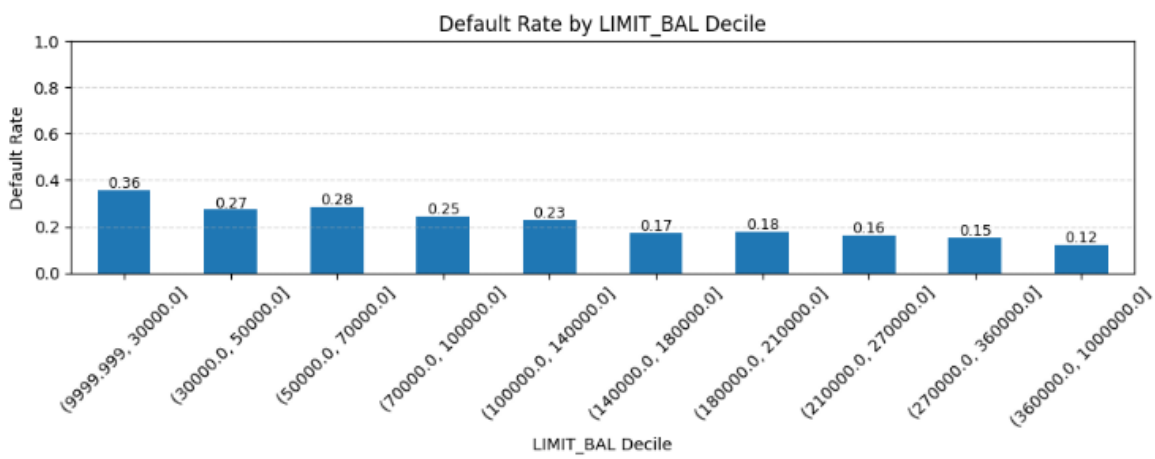

Figure 3: Default Rate by Credit Limit Decile

The distributions of financial variables such as credit limit and bill amounts are highly right-skewed, with a small number of observations taking very large values. While transformations such as logarithmic scaling could be applied, we do not pursue these in order to preserve interpretability and maintain consistent comparisons across specifications.

To understand how information is shared across predictors, we examine the correlation structure in Figure 4. Financial variables are highly correlated across time, and repayment variables exhibit strong temporal dependence, indicating substantial overlap in information.

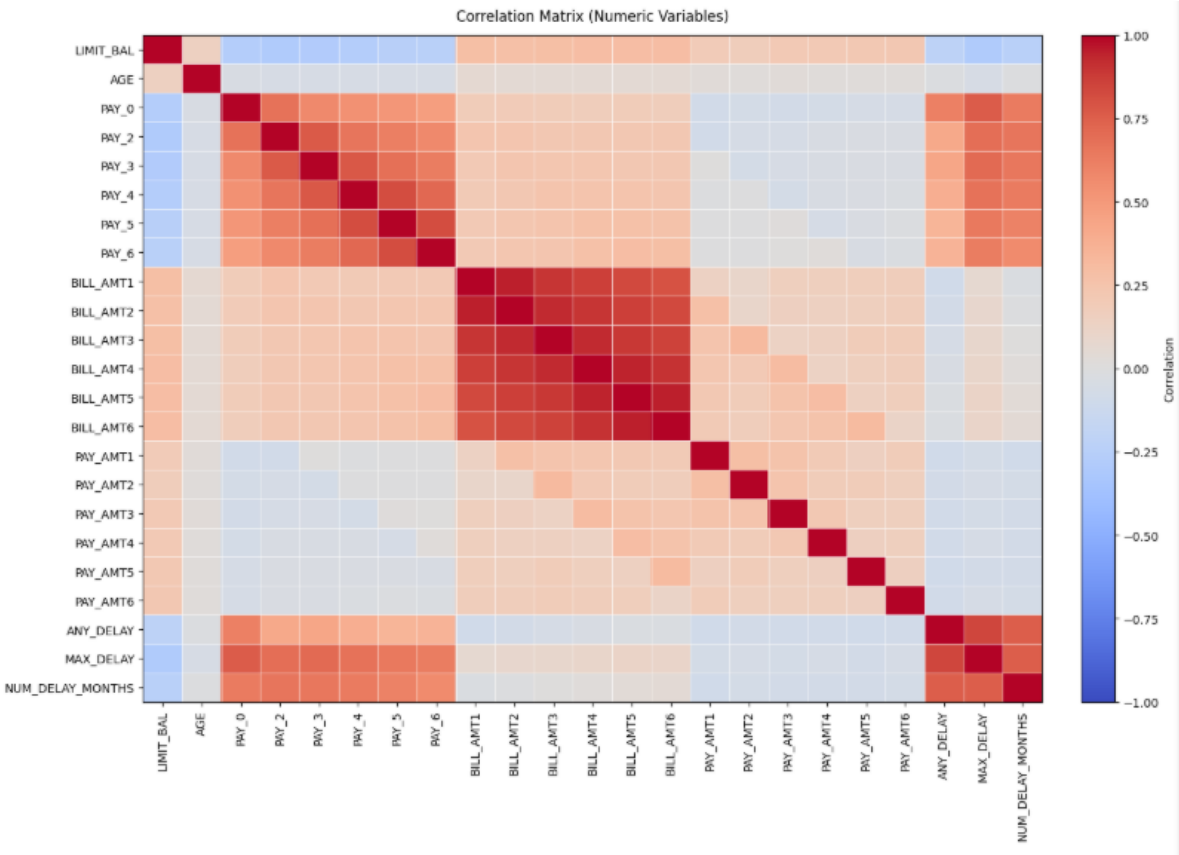

Figure 4: Correlation Matrix of Financial and Behavioral Variables

Taken together, these findings suggest that repayment behavior provides a strong and direct signal of default risk, while financial variables capture related but less immediate aspects of borrower condition.

4 Methodology

We estimate a sequence of models with varying covariate sets to evaluate how coefficient estimates and predictive performance change across specifications. In particular, we compare models based on financial variables, behavioral variables, and combinations of both, as well as models that replace the full set of repayment variables with lower-dimensional summary measures shown in (2), (3), and (4).

Ordinary least squares is used to study coefficient stability through the linear projection of the binary outcome onto the covariates. While the conditional expectation is not necessarily linear, the projection interpretation remains valid. Under the Gauss–Markov framework, OLS recovers the best linear predictor in terms of mean squared error.

We consider the following specifications. Let F_i denote financial variables, B_i denote behavioral variables, and S_i denote summary behavioral features. The models are:

$$Y_i = F_i^\top \beta_F + \varepsilon_i \tag{5}$$

$$Y_i = B_i^\top \beta_B + \varepsilon_i \tag{6}$$

$$Y_i = F_i^\top \beta_F + B_i^\top \beta_B + \varepsilon_i \tag{7}$$

$$Y_i = F_i^\top \beta_F + S_i^\top \beta_S + \varepsilon_i \tag{8}$$

For logistic regression, these correspond to:

$$\mathbb{P}(Y_i = 1 \mid X_i) = \frac{\exp(X_i^\top \beta)}{1 + \exp(X_i^\top \beta)}. \tag{9}$$

Model evaluation is conducted out-of-sample using a train-test split. This ensures that performance comparisons reflect generalization rather than in-sample fit, which is particularly important in the presence of correlated predictors. All preprocessing steps are performed using only training data to avoid data leakage.

Using the Frisch–Waugh–Lovell perspective,

$$\hat{\beta}_1 = \frac{\mathrm{Cov}(M_{X_2} X_1, M_{X_2} Y)}{\mathrm{Var}(M_{X_2} X_1)}, \tag{10}$$

highlighting that coefficients are identified from variation orthogonal to other predictors.

The classical linear model assumptions are not imposed as a basis for model validation. The objective of this analysis is not to recover a correctly specified data-generating process, but rather to compare how different sets of covariates contribute to explaining variation in the outcome and to evaluate predictive performance across specifications.

Under this interpretation, the linear model is viewed as the best linear approximation to $\mathbb{E}[Y_i \mid X_i]$, and the coefficients are understood as projection coefficients rather than structural parameters. As a result, assumptions such as homoskedasticity and normality are not required for the validity of the analysis. In particular, the binary outcome implies heteroskedasticity by construction, so deviations from constant variance are expected rather than problematic.

Instead, the primary concerns are avoiding data leakage, ensuring valid out-of-sample evaluation, and understanding how multicollinearity affects coefficient stability. The analysis therefore focuses on comparing projections across specifications and on predictive performance, rather than on validating a fully specified regression model.

5 Results

5.1 Model Performance

We begin by comparing out-of-sample predictive performance across the specifications introduced in Section 4. Figure 5 summarizes the AUC values for models based on financial variables alone, behavioral variables alone, and combinations of both.

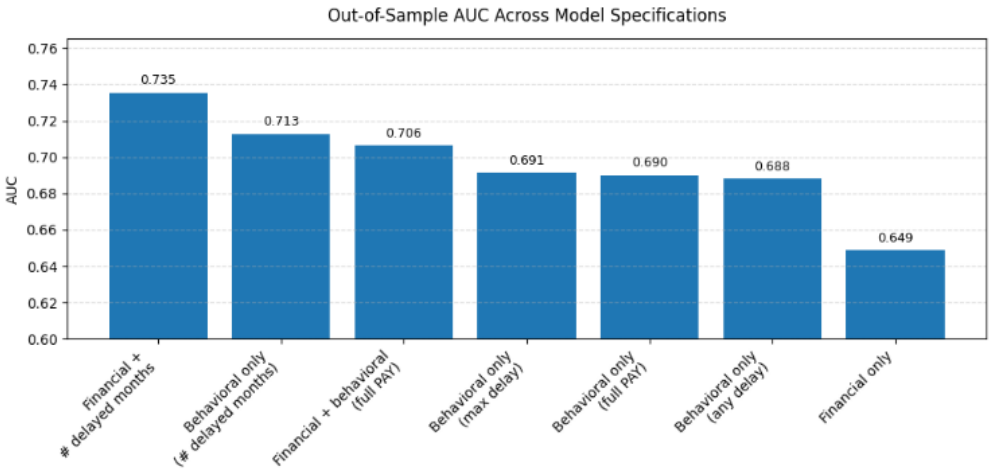

Figure 5: Out-of-Sample AUC Across Model Specifications

Financial-only models achieve AUC values in the range of approximately 0.63 to 0.65, indicating limited ability to discriminate between defaulters and non-defaulters using exposure and payment variables alone. Behavioral-only models perform materially better, with AUC values around 0.69 to 0.71, showing that delinquency behavior captures a substantial portion of the variation in default outcomes. Combined models yield only modest improvement, with the best specification reaching an AUC around 0.73 to 0.74.

This pattern is economically intuitive in a credit-risk setting. Financial variables such as balances, payments, and credit limits contain information about borrower condition, but once delinquency is directly observed, much of that information has already been realized through repayment behavior. The limited gain from adding financial variables suggests that these variables primarily act as indirect indicators of borrower risk, with much of their predictive content being realized through repayment behavior. Once delinquency is observed, financial covariates provide little additional information about near-term default risk, indicating that they function largely as leading indicators rather than independent drivers of default.

To make the ranking more transparent, Figure 6 reports a smaller set of representative specifications. This comparison isolates the main empirical result: behavioral information dominates financial information, and simple delinquency summaries can perform as well as, or better than, more granular behavioral feature sets.

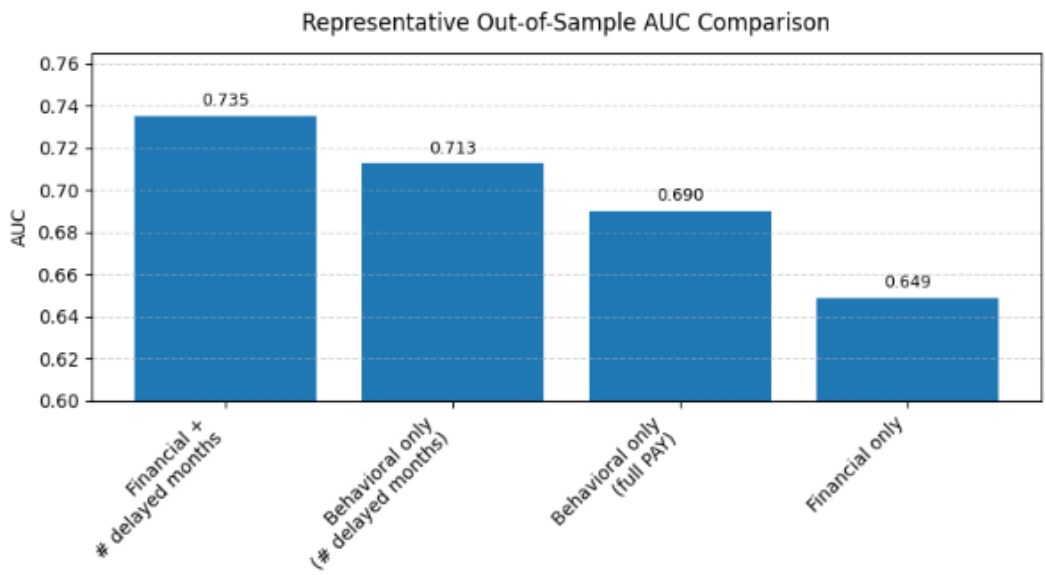

Figure 6: Representative Out-of-Sample AUC Comparison

Model	auc	log_loss	accuracy
Financial + # delayed months	0.7353	0.4600	0.8015
Behavioral only (# delayed months)	0.7128	0.4646	0.8022
Financial + behavioral (full PAY)	0.7063	0.4720	0.8092
Behavioral only (max delay)	0.6913	0.4811	0.7905
Behavioral only (full PAY)	0.6901	0.4783	0.8085
Behavioral only (any delay)	0.6879	0.4763	0.7788
Financial only	0.6487	0.5081	0.7788

Figure 7: Performance Comparison of Model Specifications

5.2 Feature Representation

A central question in the analysis is whether the full repayment history contains meaningful information beyond a smaller number of delinquency summaries. Figure 8 shows default rates as a function of the number of delayed months.

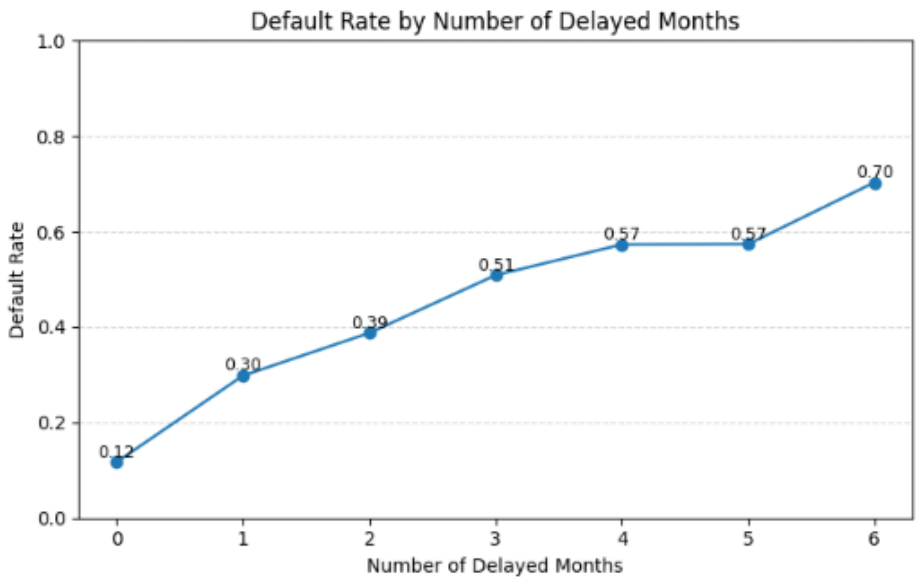

Figure 8: Default Rate by Number of Delayed Months

The relationship is strongly increasing and approximately monotonic, indicating that this summary variable captures a large share of the predictive content contained in the full repayment history. This is consistent with the model comparisons in Figure 5, where the specification using the number of delayed months performs comparably to, and in some cases better than, the full behavioral specification.

From a statistical perspective, this suggests that the repayment block is effectively low-dimensional. If B_i denotes the full vector of repayment variables and S_i denotes a lower-dimensional summary, then the empirical results indicate that

$$B_i^\top \beta_B \approx S_i^\top \beta_S$$

in predictive terms. That is, the additional components of B_i contribute limited marginal information once the main features of delinquency frequency and severity have been captured.

This finding also aligns with credit risk practice. Many underwriting and account-monitoring systems emphasize summary delinquency indicators, such as the number of missed payments or worst delinquency observed, rather than detailed month-by-month repayment sequences. The results here provide empirical support for that approach.

5.3 Coefficient Stability

We next turn to coefficient behavior across specifications. In financial-only models, several financial variables appear statistically significant. However, once behavioral variables are introduced, many of these coefficients shrink substantially or lose significance altogether. This is consistent with the view that financial variables and repayment behavior capture overlapping components of borrower risk.

The underlying issue is multicollinearity. For the OLS projection, the variance of the estimator is

$$\mathrm{Var}(\hat{\beta}) = \sigma^2 (X^\top X)^{-1}. \tag{11}$$

When predictors are highly correlated, $X^\top X$ becomes ill-conditioned, inflating the variance of the estimated coefficients and making them sensitive to the included covariates.

This interpretation can also be expressed through the Frisch–Waugh–Lovell decomposition. The coefficient on a variable X_1 conditional on controls X_2 is

$$\hat{\beta}_1 = \frac{\mathrm{Cov}(M_{X_2}X_1,\ M_{X_2}Y)}{\mathrm{Var}(M_{X_2}X_1)}, \tag{12}$$

where M_{X_2} projects onto the orthogonal complement of the span of X_2. When the variation in X_1 orthogonal to X_2 is small, the denominator becomes small and the estimate becomes unstable. In this setting, adding behavioral covariates reduces the independent variation left in many financial variables, which explains why their coefficients become smaller or less stable.

A coefficient close to zero should therefore not be interpreted as evidence that a variable is unrelated to default. Rather, it indicates that the variable provides little additional explanatory power conditional on the other predictors already in the model. In the present setting, this is best understood as a redundancy result, not a no-effect result.

To make this explicit, Figure 9 compares selected coefficients across representative specifications.

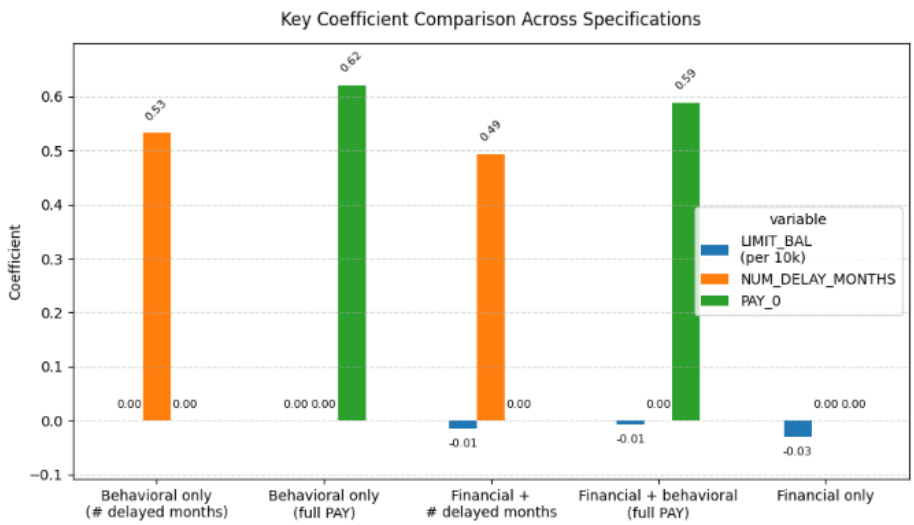

Figure 9: Key Coefficient Comparison Across Model Specifications

Figure 9 highlights the same pattern seen in the regression summaries: behavioral variables retain relatively stable explanatory power, while many financial coefficients weaken once delinquency is controlled. This reinforces the broader interpretation that financial variables act as proxies for underlying borrower risk that is more directly revealed through repayment behavior. Once delinquency variables are included, much of the independent variation in financial covariates is absorbed, leading to smaller and less stable coefficient estimates.

5.4 Delinquency Regimes

The full-sample results implicitly average over borrowers in different credit states. To examine whether variable importance depends on borrower risk regime, the sample is partitioned according to whether any delinquency is observed, and the models are re-estimated separately within non-delinquent and delinquent subsamples.

In the non-delinquent regime, behavioral variables provide little discriminatory power, with some summary behavioral specifications performing close to random classification. This is expected:

when no delinquency has yet occurred, repayment variables contain limited variation and therefore provide little usable information for ranking borrowers. In this regime, financial variables perform better, suggesting that risk must be inferred from balances, utilization, and credit capacity before repayment problems are observed.

In the delinquent regime, the pattern reverses. Behavioral variables dominate predictive performance, while financial variables provide only limited incremental value. Once repayment issues are observed, the severity and persistence of delinquency become direct measures of distress, and the informational role of financial variables becomes secondary.

Figure 10 summarizes this regime-dependent comparison.

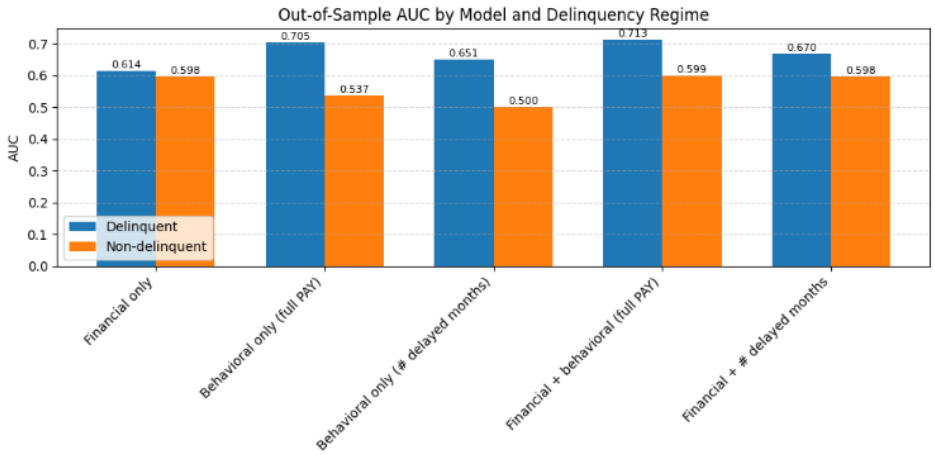

Figure 10: Out-of-Sample AUC by Model and Delinquency Regime

These results show that variable importance is state-dependent. Financial variables are more informative prior to delinquency, while behavioral variables dominate once repayment problems occur. This provides a more nuanced interpretation of the full-sample results and ties the statistical findings more closely to the credit cycle: financial variables are useful for identifying vulnerability before distress materializes, whereas behavioral variables capture realized risk once distress is observed.

5.5 Model Diagnostics

The diagnostic focus in this project differs from standard regression analysis. Rather than assessing whether a particular model satisfies classical assumptions, the goal is to understand how information is distributed across correlated predictors and how this affects coefficient interpretation and predictive performance.

Because the outcome is binary, heteroskedasticity arises mechanically under the linear projection, and deviations from constant variance do not indicate model misspecification in the usual sense. Similarly, normality assumptions are not relevant, as inference on individual coefficients is not the primary objective. Instead, the linear model is used as a tool to compare conditional projections across different covariate sets, and logistic regression is used to evaluate predictive performance.

Accordingly, the main diagnostic concern is multicollinearity, which directly affects the stability and interpretability of coefficient estimates. Other classical diagnostics are not emphasized, as they are not central to the objectives of the analysis.

To quantify multicollinearity, we compute variance inflation factors for selected predictors. Figure 10 reports the resulting values.

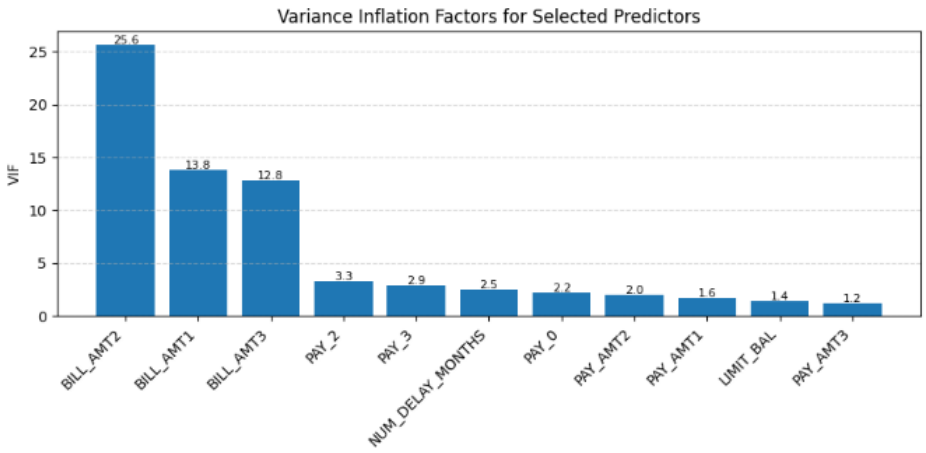

Figure 11: VIF for Selected Predictors

The VIF values confirm substantial redundancy, particularly among the financial variables and among the repayment variables across time. This supports the interpretation that unstable coefficients arise from shared explanatory content rather than idiosyncratic estimation noise.

To complement the ranking-based evaluation, Figure 12 plots the predicted probability distributions for defaulters and non-defaulters under a representative logistic specification. Specifically, we use the model that combines financial variables with the summary behavioral feature `NUM_DELAY_MONTHS`. This specification is chosen because it achieves near-maximum predictive performance while remaining parsimonious, and therefore provides a clean illustration of how the model assigns default probabilities.

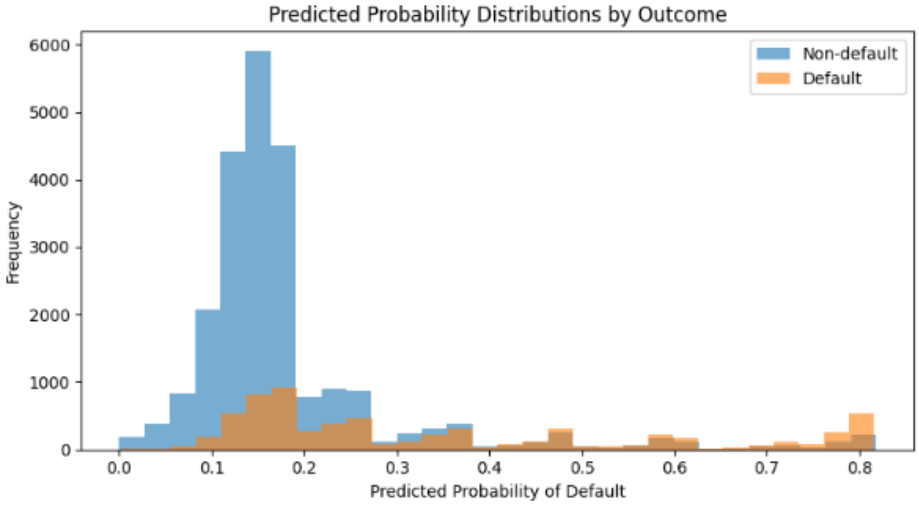

Figure 12: Predicted Probability Distributions by Outcome

The separation between the two distributions is consistent with the observed AUC values and shows that the model assigns higher probabilities to defaulters on average, even though overlap

remains substantial. This is typical in consumer credit settings, where default risk is heterogeneous and only partially observed.

6 Conclusion

This analysis shows that repayment behavior is the primary determinant of near-term default risk, while financial variables provide additional but limited explanatory power. In particular, the predictive content of repayment history can be summarized using relatively simple delinquency measures, and increasing feature dimensionality does not materially improve out-of-sample performance.

More broadly, the results highlight a limitation of regression-based interpretation in the presence of correlated predictors. When variables encode overlapping information, coefficient estimates reflect only the residual contribution of each variable after conditioning on others. As a result, variables that appear important in isolation may lose significance once more direct measures of the same underlying signal are included. In this setting, financial variables primarily act as upstream indicators of borrower risk, while repayment behavior reflects realized distress. This distinction explains both the observed dominance of behavioral variables in prediction and the instability of financial coefficients across specifications.

The regime analysis sharpens this conclusion. Financial variables are more informative before delinquency is observed, when risk must be inferred indirectly from borrower condition. Once delinquency appears, behavioral variables dominate because they measure realized distress more directly. From a credit risk perspective, these findings suggest that parsimonious delinquency-based features can deliver competitive predictive performance while offering greater interpretability and stability than more granular but redundant specifications.

As a final note on the relevance of insights, the dataset used in this analysis is drawn from credit card accounts in 2005. As a result, the absolute levels of default risk and the behavior of specific financial variables may not reflect current lending environments. However, the primary focus of this analysis is on the relative roles of behavioral and financial predictors and on the stability of coefficient estimates under correlated covariates. These structural relationships are less sensitive to the specific time period and are expected to generalize more broadly across credit settings.

7 Statement of Contribution

- Kevin Xu: Data cleaning, EDA, methodology, delinquency regimes, coefficient stability
- Chuan Qi Sun: Methodology, modeling and evaluation, model diagnostics